Exam Date & Time: 26-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

ENGINEERING MATHEMATICS - III [MAT 2121]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Solve the boundary value problem using finite difference method:
$xy'' + y = 0,\ y(1) = 1,\ y(2) = 2; h = 0.25$. (4)

1B) Solve the Poisson's equation $\nabla^2 u = -10(x^2 + y^2 + 10)$ for $0 < x < 3,\ 0 < y < 3$, given that $h = 1,\ u = 0$ on all boundary. (3)

1C) Solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$; 0 < x < 1; t > 0; h = 0.25; $u(x,0) = 100(x - x^2); u(0,t) = u(1,t) = 0$ up to four time steps by taking $\lambda = \frac{1}{2}$. (3)

2A) Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, 0 < x < 1, t > 0, u(x,0) = 100 sin\, \pi x,\ \frac{\partial u}{\partial t}(x,0) = 0, u(0,t) = u(1,t) = 0,\ h = \frac{1}{4}$, for four time steps. (4)

2B) Find the Fourier series up to the first harmonics for y = f(x) in $(0, 2\pi)$ defined by the table of values given below. (3)

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$
y	1	1.4	1.9	1.7	1.5	1.2

2C) Obtain half range sine series for $f(x) = x \quad 0 \le x \le \pi.$ in the interval (3)

3A) Obtain the Fourier series expansion for the function

$f(x) = x(2\pi - x), 0 < x < 2\pi,\ f(x + 2\pi) = f(x).$ (4)

3B) (3)

Find the Fourier transform of the function $f(x) = \begin{cases} 1, & for\ |x| \le 1 \\ 0, & for\ |x| > 1 \end{cases}$.

3C) Find the directional derivative of $P = 4e^{2x-y+z}$ at $(1,1,-1)$ in a direction towards the point $(-3,5,6)$. (3)

4A) Find the acute angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at $(1,-2,1)$. (4)

4B) Evaluate using Stokes theorem, $\oiint (\nabla \times \vec{A}) \cdot \hat{n}\, ds$ where $\vec{A} = (x^2 + y - 4)\hat{\imath} + 3xy\hat{\jmath} + (2xz + z^2)\widehat{k}$ and S is the surface of the hemisphere $x^2 + y^2 + z^2 = 16$ above the xy-plane. (3)

4C) Suppose a force field is given by $\vec{F} = (2x - y + z)\hat{\imath} + (x + y - z^2)\hat{\jmath} + (3x - 2y + 4z)\hat{k}$. Find the work done in moving a particle once around a circle C in the xy-plane with its centre at the origin and a radius of 3. (3)

5A) Verify Green's theorem in plane for $\int_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$ where C is the boundary of the region defined by $x = 0, y = 0\ and\ x + y = 1$. (4)

5B) Solve $u_x + u_y = 2(x + y)u$, by separating the variables. (3)

5C) Solve the partial differential equation $\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$ using the transformation $v = x + t,\ \ w = x - t$. (3)

-----End-----